

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

de Broglie wavelength of a particle is calculated using the formula $\lambda = h/p$, where λ is the wavelength and h is Planck's constant . Given the kinetic energy of the electron ($E_k = 1\text{ eV}$), we need to calculate its momentum p first .

To calculate the momentum , we'll need to convert the kinetic energy from eV to Joules , as Planck's constant (h) is in units of Js .

$1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

1 eV kinetic energy is ($1.602 \times 10^{-19}\text{ J} = 1.602 \times 10^{-19}\text{ J}$)

so for 1 eV energy 1 eV kinetic energy = $1.602 \times 10^{-19}\text{ J}$

$|p| = \sqrt{2 m \times 1\text{ eV kinetic energy}} = \sqrt{2 \times 1.602 \times 10^{-19}\text{ J}} = 4.899 \times 10^{-10}\text{ kg m/s}$

$\lambda = h/p$

Planck constant $h = 6.626 \times 10^{-34}\text{ J m}$

$\lambda = (6.626 \times 10^{-34}\text{ J m}) / 4.899 \times 10^{-10}\text{ dm}$

$\lambda = 1.34 \times 10^{-25} = 1.34 \times 10^{-25}\text{ measure}$